



V₂CT_x MXene — Single-/Few-Layer Powder

Research-grade two-dimensional transition-metal carbide powder

This delaminated V₂CT_x MXene powder is produced from V₂AlC by HF etching followed by intercalation and delamination. The material has a single-/few-layer structure of no more than five layers and nanoscale lateral dimensions. Surface terminations include -O, -OH, -F, and -Cl.

KEY FEATURES

- Delaminated single-/few-layer morphology (≤ 5 layers)
- Nanoscale nominal lateral size
- Typical conductivity of 1500 ± 500 S/cm
- Mixed -O/-OH/-F/-Cl surface terminations

POTENTIAL RESEARCH APPLICATIONS

- Electrochemical energy-storage and metal-sulfur battery research
- Electrocatalysis and conductive-composite studies
- Two-dimensional materials and interface research

STORAGE & HANDLING

Keep dry in a tightly closed container. Vacuum or inert-atmosphere storage is recommended. Minimize exposure to moisture and air; use appropriate controls for fine powders.

QUALITY / DOCUMENTATION

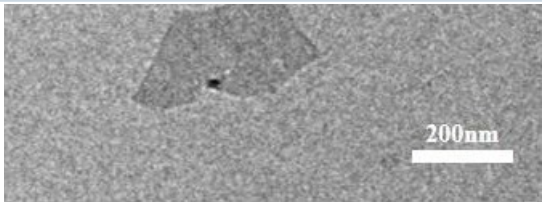
Surface termination distributions vary with synthesis and lot. Quantitative surface composition can be discussed when required.

TECHNICAL SPECIFICATIONS

PROPERTY	TYPICAL / REPRESENTATIVE VALUE
Material	V ₂ CT _x MXene
Precursor	V ₂ AlC
Product form	Delaminated single-/few-layer powder
Preparation	HF etching followed by intercalation / delamination
Layer structure / count	≤ 5 layers
Nominal lateral size	200–1000 nm
Typical conductivity	1500 ± 500 S/cm
Surface terminations	-O, -OH, -F, and -Cl
Appearance	Dark powder (descriptive only)
Recommended storage	Dry; vacuum or inert atmosphere

Conductivity values are representative technical values. Measurement results can vary with sample preparation, packing, film preparation and measurement geometry.

REPRESENTATIVE CHARACTERIZATION

	Single-/few-layer V ₂ CT _x — TEM. Representative characterization; not lot-specific.
	Single-/few-layer V ₂ CT _x — XRD. Representative characterization; not lot-specific. Non-data annotation removed for presentation; diffraction trace and axes otherwise unchanged.

Product positioning: Supplied and technically supported by WiniGen Materials for research and development use.

WINIGEN MATERIALS
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V2CTX-SFL-TDS-001

Representative technical values. Properties can vary with synthesis and lot. Lot-specific documentation and additional analytical details are available subject to review.